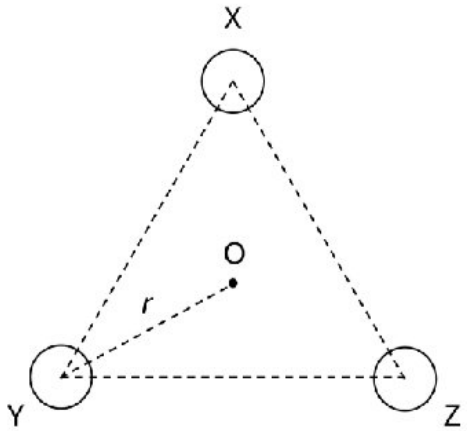
Use the following diagram to answer questions 8 and 9.

Three planets, X, Y and Z have masses *M*, *M* and *2M* respectively, are arranged in a triangle and are equidistant from each other.



8 What is the magnitude and direction of the gravitational field strength at point O?

	magnitude	**direction**
A	0	no direction
B	$\frac{GM}{r^2}$	towards X
C	$\frac{GM}{r^2}$	towards Z
D	$\frac{G\left(2M-\sqrt{3}M\right)}{r^2}$	towards Z